MAIZE YIELD PREDICTION

Uasin Gishu County, Kenya

RESEARCH QUESTION

Can annual maize yield (tonnes per hectare) in Uasin Gishu County be predicted using seasonal weather, soil, and seed management data?

1. Executive Summary

3.65 t/ha County Average Yield *2012–2023*	**4.26 t/ha** Best Year on Record *2018*	**2.82 t/ha** Worst Year on Record *2012 — MLN outbreak*	**~6.5 t/ha** Yield Gap vs Potential *Seed potential: 9–11 t/ha*

Uasin Gishu County is one of Kenya's most productive maize-growing regions. Despite fertile land and access to improved hybrid seed varieties, farmers consistently harvest far below what those seeds are capable of producing. This study investigates whether seasonal weather patterns explain year-to-year yield variation, and builds a machine learning model that can predict yield from those conditions before harvest.

Using 12 years of data (2012–2023), five datasets covering yields, weather, soil properties, and seed varieties were combined into a single analytical table. Three central findings emerged:

1. **Rainfall timing matters more than total rainfall volume.**

   2023 had over 2,000 mm of annual rain and produced 4.04 t/ha. 2020 received even more — 2,371 mm — yet produced only 3.07 t/ha because the bulk of that rain arrived outside the planting window.

2. **Conditions before planting shape the following harvest.**

   Humidity and soil conditions in the months before planting (September–February) were stronger predictors of yield than conditions during the planting season itself.

3. **The yield gap is a soil problem, not a weather problem.**

   The ~6.5 t/ha gap between the county average (3.65 t/ha) and seed potential (9–11 t/ha) cannot be closed by better weather. Soil acidity at pH 5.7 — below the optimal range of 6.0–7.0 — is the primary barrier. Soil liming is the intervention that weather management cannot replace.

2. Background and Problem Statement

Uasin Gishu County sits in western Kenya at an average elevation of 1,900 metres above sea level. It is one of Kenya's most important maize-producing counties, with a bimodal rainfall pattern: Long Rains from March to May (the main planting season) and Short Rains from September to November.

Despite access to improved hybrid seed varieties with a genetic yield potential of 9–11 tonnes per hectare, the county's average actual yield over 2012–2023 was just 3.65 t/ha — roughly one-third of what the seeds are capable of producing. The gap between what is harvested and what could be harvested — approximately 6.5 t/ha — represents a significant lost opportunity for food security and farm income.

This analysis pursues two questions: first, how much of the year-to-year variation in yield can be explained by seasonal weather? Second, why does a structural gap of ~6.5 t/ha persist regardless of annual weather conditions? Answering both questions is necessary to direct the right interventions to the right causes.

3. Data Sources

Five separate datasets from four organisations were assembled and merged into a single 12-year analytical table. The table below summarises each source.

Dataset	Source	Format	Coverage	What It Contains
Maize yield	Ministry of Agriculture & Livestock, Kenya	CSV	2012–2023	Annual yield in tonnes per hectare
Area planted	Ministry of Agriculture & Livestock, Kenya	CSV	2012–2023	Hectares planted each year
Weather data	NASA POWER Satellite Climate Archive	Excel	2012–2023	Monthly rainfall, temperature, humidity, soil wetness, solar radiation
Soil properties	Lomurut Survey 2014 / Purdue University	CSV	Static (2014)	pH, nitrogen content, soil classification across four field sites
Seed varieties	CIMMYT / Tegemeo Institute	Excel	2012–2023	Variety used per year, yield potential range, disease risk indicator

After merging all five sources on the Year column, the resulting master table contained 12 rows (one per year) and 38 columns, with zero missing values across all fields.

4. Methodology

4.1 Data Cleaning and Preparation

Maize Yield and Area Planted

Both files shared the same structure. Only two columns were retained from each: Year and Value. The value column was renamed to Yield_Tonnes_Ha in the yield file and Area_Planted_Ha in the area file. Both datasets were sorted by year in ascending order.

Seed Variety Data

The variety name and yield potential range were retained from each year's records. Two additional calculated columns were then derived:

New Column	Formula	What It Records
MLN_Risk	1 if the Year falls in 2011, 2012, or 2013; otherwise 0	Flags years affected by the Maize Lethal Necrosis disease outbreak that caused widespread crop losses across Kenya
Variety_Code	H614 = 0, H6213 = 1, H6218 = 2, H629 = 3, DK8031 = 4	A numeric label for each seed variety, allowing the model to handle variety names as numbers if needed

Soil Data

The soil dataset contained 18 rows representing multiple depth layers from four sampling sites across the county. Only the topsoil layer was retained (depth code HONU = 1, which corresponds to the root zone most relevant to crop nutrition). The county-wide median was then calculated across all four sites, producing three fixed values:

Soil Property	Value	Optimal Range for Maize	Significance for This Study
Soil pH	5.7	6.0 – 7.0	Below pH 6.0, aluminium in the soil becomes toxic to maize roots — this is the primary explanation for the structural yield gap
Total Nitrogen	3.8 g/kg	2.0 – 5.0 g/kg	Within the acceptable range; not a limiting factor
Soil Class	GLe (Gleysol)	N/A	Waterlogging-prone soil type; poor natural drainage in wet years

Because these soil values are the same for every year in the dataset, they cannot statistically explain why yield differs from year to year. They are therefore excluded from the prediction model. However, they are central to understanding the structural yield gap discussed in Section 5.

4.2 Weather Feature Engineering

The NASA POWER dataset arrived in wide format: each row represented one year-and-climate-variable combination, with separate columns for each of the 12 months (JAN through DEC) and an annual summary column (ANN). Seven raw climate variables were extracted:

NASA Code	Plain Name	Unit
PRECTOTCORR_SUM	Rainfall	mm per month
T2M	Average temperature at 2 metres height	°C
T2M_MAX	Maximum daily temperature	°C
T2M_MIN	Minimum daily temperature	°C
RH2M	Relative humidity at 2 metres height	Percent
GWETTOP	Surface soil wetness	Scale 0 (dry) to 1 (saturated)
ALLSKY_SFC_SW_DWN	Solar radiation (sunlight energy reaching the surface)	MJ/m² per day

Why Seasons Rather Than Annual Averages?

Annual weather averages mask when conditions occurred, and timing is fundamental to crop biology. Maize germinates, grows, pollinates, and fills grain at specific calendar periods. Rain arriving during the wrong month — even in large volumes — does not support the same growth as the same rain arriving at the right time.

The year was therefore divided into four agronomic seasons defined by the local agricultural calendar:

Season	Months	Agricultural Significance
Long Rains	March, April, May	The main planting and growing season. The most critical window for germination, vegetative growth, and pollination.
Off Season	June, July, August	Normally the dry period. Unexpected rain here causes disease pressure and disrupts soil preparation for the next season.
Short Rains	September, October, November	The second planting season. Conditions in this period determine soil health going into the following Long Rains.
Dry Season	December, January, February	Land preparation period. Excess rain at this stage degrades soil structure and drainage before planting begins.

For each raw climate variable and each season, a single summary figure was calculated:

- Rainfall → Season total (sum of the three monthly values in mm)
- Temperature, Humidity, Soil Wetness, Solar Radiation → Season average (mean of the three monthly values)

This produced 21 seasonal weather features, such as Rain_LongRains_mm, Temp_ShortRains_C, Humidity_ShortRains_pct, and SoilWetness_OffSeason, spanning all four seasons and all relevant climate variables.

Derived Variables: Five Additional Computed Features

Five further variables were computed from the seasonal summaries. Each captures an agronomic concept that the raw numbers alone do not directly express. Plain-language formulas are shown below.

Variable	Formula	What It Captures	Why It Matters for Maize
Rain_LongRains_Fraction	Long Rains total mm ÷ Annual total mm	The proportion of the year's rainfall that fell during the planting season	A high fraction means rain arrived when the crop needed it most. This became the single strongest predictor of yield (r = +0.61).
Heat_Stress_LongRains	1 if maximum temperature during Long Rains > 30°C; otherwise 0	A binary flag for years when temperatures exceeded the safe limit for pollination	Maize pollen dies above 30°C. This flag marks years when that biological threshold was crossed.
Temp_Range_Annual_C	Highest temperature recorded in the year − Lowest temperature recorded	The full annual swing between the hottest and coldest days	Large temperature swings indicate climate instability and can signal stress events not visible in averages.
Drought_LongRains	Long Rains rainfall mm ÷ (Long Rains average temperature °C + 1)	A combined heat-and-dryness stress index for the growing season	Low values (little rain, high heat) signal potential drought stress during the critical growth period.
Moisture_Effectiveness_LongRains	Long Rains rainfall mm × Long Rains average soil wetness	How effectively planting-season rain is absorbed into the soil	Rain falling on already-saturated soil runs off rather than being absorbed. This variable captures that distinction.

After all cleaning, merging, and feature engineering steps, the master table contained 12 rows and 38 columns.

4.3 Feature Selection: From 38 Variables to 6

With only 12 years of data, using all 38 variables in the model would cause overfitting — the model would memorise the 12 specific training years rather than learn genuine weather-to-yield patterns that generalise to new seasons. A disciplined selection process was applied using three criteria applied together:

1. Statistical strength: The Pearson correlation coefficient (r) was calculated between each variable and yield. Priority was given to variables with an absolute correlation of 0.45 or above.
2. Agronomic logic: Each selected variable must have a documented, physically plausible reason for affecting maize growth. Variables that were statistically correlated but had no clear agronomic explanation were treated with caution.
3. Low redundancy: Where two variables were strongly correlated with each other (measuring nearly the same underlying condition), only the stronger predictor of yield was retained. Including both would give that signal disproportionate weight in the model.

Variables excluded and the reason for their exclusion:

- Soil pH, Nitrogen, Fertiliser rate — identical value every year; mathematically impossible to explain year-to-year yield variation with a constant.
- Year — a time trend is not a weather relationship and would not generalise to future seasons.
- Area planted — reflects farmer expectations, not yield-driving conditions; correlation r = +0.06.

- Variety code and MLN Risk — near-zero predictive signal once weather variables are included; too few observations per variety to reliably separate variety effect from weather effect.
- Heat Stress flag, Temperature Range, Drought Index, Moisture Effectiveness — weaker signal in testing than the simpler seasonal variables from which they were derived.

The six variables retained for the final model are shown below.

Variable	Plain Description	Correlation with Yield	Agronomic Reason for Inclusion
Rain_LongRains_Fraction	Share of annual rain that fell in Mar–May	r = +0.61 (strongest predictor)	Rain concentrated in the planting window directly supports germination and early growth. Rain spread across other seasons does not compensate.
Humidity_ShortRains_pct	Average humidity Sep–Nov (%)	r = −0.60	High humidity in the months before planting increases fungal disease pressure and creates waterlogging conditions that persist into the growing season.
Rain_DrySeason_mm	Total rainfall Dec–Feb (mm)	r = −0.59	Excess rain during land preparation degrades soil structure, reduces drainage capacity, and prevents timely field preparation before planting.
Temp_ShortRains_C	Average temperature Sep–Nov (°C)	r = +0.52	Warmer temperatures during the pre-planting soil recovery period support better soil biological activity and root conditions going into the Long Rains.
SoilWetness_ShortRains	Soil saturation Sep–Nov (0–1 scale)	r = −0.49	Saturated soils before the growing season restrict root development, reduce available oxygen in the soil, and impair drainage during early planting.
Rain_OffSeason_mm	Total rainfall Jun–Aug (mm)	r = −0.49	Unexpected rain during the normally dry off-season disrupts soil preparation activities and creates conditions favourable for fungal disease.

4.4 Model Training

Two machine learning models were trained and their performance compared. Using two models rather than one provides a cross-check: if both models agree on a prediction, confidence in that prediction is higher.

Model 1: Random Forest

A Random Forest constructs a large number of decision trees, each trained on a randomly selected subset of the data and a randomly selected subset of the predictor variables. The individual tree predictions are then averaged into a single final estimate. This averaging process smooths out errors that any single tree would make on unusual years.

To prevent the trees from simply memorising the 12 training years, three constraints were applied: a maximum tree depth of 3 levels, a minimum of 3 data points required at every leaf node, and consideration of only the square root of available features at each split point. In total, 500 trees were grown.

Model 2: Support Vector Regression (SVR)

SVR finds the best-fitting curve through the data while deliberately tolerating small prediction errors (within a margin of 0.1 t/ha). This makes it less sensitive to any single unusual year in the training data. A radial basis function kernel was used, which allows the model to learn curved (non-linear) relationships between weather inputs and yield. Regularisation was set to C = 1.0, meaning the model accepts some imperfection on training data rather than fitting every point exactly.

Feature Scaling

Before training, all six predictor variables were standardised using a Standard Scaler: each variable was transformed to have a mean of zero and a standard deviation of one. This is required for the SVR algorithm, which is sensitive to differences in the scale of input variables. The same scaling transformation was saved and applied identically to all new inputs at prediction time.

4.5 Validation: Leave-One-Out Cross-Validation

With only 12 years of data, standard approaches such as an 80/20 train-test split are not statistically reliable — any single split might produce a result that is unusually favourable or unfavourable due to which years happen to fall in each group. Leave-One-Out Cross-Validation (LOO-CV) was used instead, which is the most honest validation method available for a dataset of this size.

How Leave-One-Out Cross-Validation works:

4. Hold out Year 1. Train the model on the remaining 11 years. Use the trained model to predict Year 1.
5. Hold out Year 2. Train the model on the remaining 11 years. Predict Year 2.
6. Repeat this process for all 12 years, holding out one year at a time.

The result is 12 genuine out-of-sample predictions — each made by a model that had never seen that year during its training. All performance metrics reported in this analysis are calculated from these 12 LOO predictions only. No metric is derived from in-sample (training) performance, which would overstate accuracy.

Three metrics are used to evaluate the model:

- RMSE (Root Mean Squared Error): The average prediction error in t/ha, giving more weight to large errors. Lower is better.
- MAE (Mean Absolute Error): The average absolute error in t/ha. Less sensitive to occasional large errors than RMSE.
- R^2 (R-squared): The proportion of year-to-year yield variation explained by the model. A value of 1.0 is perfect; 0.0 means the model does no better than always predicting the county average; negative means it is worse than that baseline.

5. Key Findings

Finding 1 — Rainfall Timing Matters More Than Total Volume

The single strongest predictor of yield was not total annual rainfall, but when the rain arrived. The variable Rain_LongRains_Fraction — the proportion of annual rainfall falling during the main planting season (March–May) — showed a correlation of r = +0.61 with yield.

The clearest demonstration is 2020. That year recorded the highest annual rainfall in the entire 12-year dataset at 2,371 mm, yet produced only 3.07 t/ha — the third lowest yield on record. The reason: only 34% of that rain fell during the March–May planting window. The remaining two-thirds arrived at times that caused waterlogging rather than supporting crop growth.

By contrast, 2018 received 1,885 mm in total — measurably less than 2020 — but 53% of it fell during the planting season. Yield reached 4.26 t/ha, the highest in the study period.

Finding 2 — Pre-Planting Conditions Shape the Following Harvest

Four of the six model variables describe conditions in the months before the main planting season rather than during it: Short Rains humidity and soil wetness (September–November), dry-season rainfall (December–February), and off-season rainfall (June–August). The combined predictive weight of these pre-season variables exceeded that of the planting-season variables themselves.

High humidity before planting elevates fungal disease pressure. Excess rainfall during land preparation degrades soil structure and drainage capacity. Both conditions constrain the crop before it has been planted. This finding has practical implications: monitoring seasonal conditions from September onwards — well before planting decisions are made — provides an early signal of likely harvest outcomes.

Finding 3 — The Yield Gap Is a Soil Problem, Not a Weather Problem

The county's average yield of 3.65 t/ha sits approximately 60% below the 9–11 t/ha genetic potential of the seed varieties in use. Even in the best weather year on record (2018), yield reached only 4.26 t/ha. The model confirms that seasonal weather explains only 30% of year-to-year yield variation, leaving 70% unexplained.

The primary cause of the persistent gap is soil acidity. The county's topsoil pH of 5.7 falls below the 6.0–7.0 range that maize requires for healthy root function. At pH 5.7, aluminium compounds in the soil become soluble and toxic to maize roots, physically blocking the uptake of phosphorus and other nutrients — even when fertiliser is applied at the full recommended rate of 124 kg/ha.

> Practical implication: Agricultural lime application to raise soil pH from 5.7 toward 6.5 is the intervention most likely to close the yield gap. Evidence from similar soils across western Kenya consistently shows yield improvements of 30–50% following liming. No combination of weather management, better seeds, or fertiliser can achieve comparable results while soil acidity remains at current levels.

Finding 4 — The Seed Variety Effect Cannot Be Reliably Measured

Five different seed varieties were grown across the 12 years of the study. The best year on record (2018) used the variety H6213, which appears in only one season in the dataset. With so few observations per variety, it is not statistically possible to separate the variety's contribution to yield from the specific weather conditions of those same years. Variety was therefore excluded from the model to avoid attributing to seed what may have been caused by weather, or vice versa.

6. Year-by-Year Data

The table below records the actual yield alongside key weather conditions and the seed variety for each year in the study period.

Year	Yield (t/ha)	Long Rains (mm)	Annual Rain (mm)	Variety	Notable Event or Condition
2012	2.82	569	1,712	DK8031	Lowest yield on record — MLN disease outbreak
2013	4.10	608	1,418	DK8031	Strong yield despite continued MLN risk
2014	3.93	319	1,142	H629	Lowest total annual rainfall in study period
2015	3.89	633	1,708	H6218	Average season; no notable extremes
2016	3.95	621	1,444	DK8031	Good planting-season rainfall timing
2017	3.31	479	1,655	H629	Below average; poor seasonal rainfall distribution
2018	4.26	991	1,885	H6213	Best year on record; 53% of rain in planting window
2019	3.24	301	1,700	H629	Very low Long Rains despite moderate annual total
2020	3.07	810	2,371	DK8031	Record annual rain; waterlogging — only 34% in planting season
2021	3.68	566	1,504	H629	Slightly above average; stable conditions
2022	3.50	389	1,519	H6218	Below-average Long Rains; dry planting season
2023	4.04	712	2,013	DK8031	Good planting-season timing despite high annual total

7. Model Performance

All performance figures are derived from Leave-One-Out cross-validation. Every prediction listed was produced by a model that had not been trained on that year.

0.364 t/ha Average Prediction Error *Random Forest LOO RMSE*	**30%** Variance Explained *LOO R^2 = 0.304*	**10%** Error vs County Average *County mean = 3.65 t/ha*	**≈0** Model Bias *−0.007 t/ha*

Model	LOO RMSE (t/ha)	LOO MAE (t/ha)	LOO R^2	Train RMSE	Outcome
Random Forest	0.364	0.283	0.304	0.246	Best model — selected for interactive dashboard
Support Vector Regression	0.397	0.310	0.174	0.163	Secondary estimate — shown alongside RF in dashboard

The Random Forest outperformed the SVR on all three LOO metrics. A prediction error of 0.36 t/ha against a county mean of 3.65 t/ha (10% of the average) represents genuine predictive value, particularly given that the entire range of observed yields over 12 years spans only 1.44 t/ha (from 2.82 to 4.26 t/ha).

7.1 Actual vs Predicted — All 12 Years

Year	Actual (t/ha)	RF Predicted	SVR Predicted	RF Error	Assessment
2012	2.82	3.55	3.55	+0.73	Largest over-prediction — MLN disease not included in model
2013	4.10	3.81	3.73	−0.28	Good prediction
2014	3.93	3.76	3.79	−0.18	Good prediction
2015	3.89	3.75	3.87	−0.14	Good prediction
2016	3.95	3.89	4.04	−0.06	Excellent prediction
2017	3.31	3.42	3.28	+0.11	Good prediction
2018	4.26	3.53	3.57	−0.73	Largest under-prediction — exceptional year with rare seed variety
2019	3.24	3.55	3.57	+0.31	Reasonable prediction
2020	3.07	3.45	3.72	+0.39	Waterlogging effect partially captured
2021	3.68	3.80	3.76	+0.12	Good prediction
2022	3.50	3.50	3.23	~0.00	Excellent prediction
2023	4.04	3.69	3.72	−0.35	Good prediction

The two largest prediction errors occurred in 2012 (MLN disease outbreak not represented by any weather variable) and 2018 (an unusual combination of excellent planting-season rain and a seed variety seen only once in the dataset). Both errors have identifiable, explainable causes rather than representing random model failure.

8. Variable Importance

To measure each variable's contribution to model accuracy, permutation importance was calculated. The fully trained model was used, then each variable was shuffled one at a time to break its relationship with yield. The increase in prediction error caused by shuffling that variable shows how much the model depended on it. A larger increase means greater importance.

Rank	Variable	Importance Score	Correlation	What It Captures
1	Humidity — Short Rains (Sep–Nov %)	0.0290	r = −0.60	Disease and waterlogging risk in the pre-planting period
2	Rainfall — Dry Season (Dec–Feb mm)	0.0287	r = −0.59	Soil disruption and drainage impairment during land preparation
3	Long Rains Rainfall Fraction	0.0249	r = +0.61	Whether annual rainfall was concentrated in the planting window
4	Temperature — Short Rains (Sep–Nov °C)	0.0132	r = +0.52	Soil recovery and biological conditions before the growing season
5	Rainfall — Off Season (Jun–Aug mm)	0.0073	r = −0.49	Unexpected rainfall disrupting the normal dry-period soil preparation
6	Soil Wetness — Short Rains (Sep–Nov)	0.0041	r = −0.49	Soil saturation levels restricting root development before planting

> All six variables carry positive importance scores, meaning removing any one of them would make predictions less accurate. The model relies on a balanced combination of pre-planting and in-season conditions rather than depending heavily on a single variable. Notably, three of the top four variables describe conditions in the months before planting begins.

9. Strengths and Limitations

9.1 Strengths

Strength	Rating	Description
Seasonal weather design	★★★★★	Splitting the year into four agronomic seasons captures when conditions occurred, not merely whether they occurred. The 2020 anomaly — record rain producing below-average yield — is correctly identified as a timing problem only when seasonal structure is applied.
Honest validation	★★★★★	Every performance metric reported in this study comes from Leave-One-Out cross-validation. No metric is derived from predicting data the model was trained on, which would overstate accuracy.
No data leakage	★★★★★	The yield variable and all variables mathematically derived from it were excluded from model inputs. Data leakage is a common and consequential error in machine learning studies; it does not occur here.
Variable selection	★★★★☆	Six variables were selected from 38 candidates using three criteria applied together: statistical correlation, agronomic logic, and redundancy checks. Each has a documented physical reason for affecting maize yield.
Model choice	★★★★☆	Random Forest and SVR are well-suited algorithms for this data size and structure. Both handle non-linear relationships and correlated predictors without requiring large sample sizes.

9.2 Limitations

Limitation	Rating	Description
Small dataset (n = 12)	★★☆☆☆	The fundamental constraint on this analysis. With only 12 data points, even a strong correlation of r = 0.60 carries an uncertainty range of approximately ±0.25. All results should be treated as directional rather than precise. The only resolution is additional historical data.
Seasonal averages hide extreme events	★★★☆☆	A single week of 35°C heat during pollination can destroy a maize crop; a 3-month seasonal average temperature would not flag this event. Monthly or daily data would allow the model to capture intra-season extremes.
Disease and pest pressure not included	★★★☆☆	The 2012–2013 MLN outbreak caused the two largest prediction errors in the model. A disease surveillance variable would reduce these errors, but the necessary data was not available for this study.
Overall confidence for operational use	★★★☆☆	Results are valid for understanding key yield drivers and running what-if scenarios. They are not suitable for precise operational yield forecasting without a larger dataset to reduce uncertainty.

10. Recommendations

Three practical recommendations follow directly from the analysis findings, targeted at farmers, extension officers, and county agricultural planners in Uasin Gishu.

Recommendation 1: Make planting and investment decisions based on rainfall timing, not just totals

The analysis consistently shows that a season where 40% or more of annual rainfall falls during the March–May window is associated with above-average yields. County extension services should communicate not just seasonal rainfall forecasts, but the expected distribution of that rainfall across the year. When early indicators suggest strong Long Rains (high planting-season fraction), farmers can invest more confidently in seeds and inputs. When the planting window looks unfavourable, risk-reduction strategies — adjusted planting dates, water conservation measures, or diversified cropping — become more justified.

Recommendation 2: Prioritise soil liming as the highest-leverage intervention

The persistent yield gap of ~6.5 t/ha cannot be closed through weather management, better seed varieties, or increased fertiliser application alone. The soil acidity constraint must be addressed first. Raising soil pH from 5.7 toward the 6.0–7.0 optimal range through agricultural lime application at approximately 1–2 tonnes per hectare would eliminate aluminium toxicity, allowing roots to absorb the fertiliser nutrients already being applied. Evidence from comparable soils in western Kenya consistently documents yield improvements of 30–50% following liming. County agricultural programmes should treat lime subsidies or distribution as a priority investment.

Recommendation 3: Begin monitoring pre-planting conditions from September onwards

Three of the top four predictive variables in the model describe conditions during September–February, well before the main planting season opens in March. County extension services and farmers should monitor Short Rains humidity and soil moisture from September. Abnormally high humidity (above 75%) in this period signals elevated disease and waterlogging risk for the following harvest; excess December–February rainfall signals likely soil drainage problems. Early awareness of these conditions allows timely drainage management, adjusted soil preparation schedules, and proactive disease prevention before planting begins.

11. Next Steps for Future Research

- Extend the dataset to 20 or more years to reduce statistical uncertainty in all correlation estimates and model performance metrics.
- Add disease surveillance indicators — particularly for Maize Lethal Necrosis and Fall Armyworm — as model variables to improve accuracy in outbreak years.
- Transition from seasonal summaries to monthly or weekly weather data to capture within-season extreme events such as heat spikes during pollination or flooding at crop establishment.
- Test the model's weather-yield relationships on adjacent counties (Elgeyo-Marakwet, Trans Nzoia, Nandi) to assess geographic transferability and identify county-specific adjustments.
- Incorporate sub-county field-level data to account for topographic and soil variation across the county's range of elevations and soil types.
- Conduct a controlled liming trial across a representative sample of farms to quantify the yield response to pH correction under current seed varieties and fertiliser practice.

12. Interactive Dashboard

An interactive web dashboard was developed to make the trained model accessible to users without technical backgrounds. Seasonal weather conditions can be entered using simple sliders, and an instant yield prediction is returned alongside a ±0.36 t/ha uncertainty range derived from cross-validation.

Resource	Link
Live Interactive Dashboard	https://maize-yieldprediction.streamlit.app
Full Code and Data (GitHub)	https://github.com/Caleb403/maize-yield-prediction

How to Use the Dashboard

7. Open either link above in any web browser — no software installation is required.
8. Use the sliders in the left panel to enter seasonal rainfall, temperature, humidity, and soil wetness values for the season of interest.
9. The yield prediction updates instantly, displaying the estimated yield alongside the uncertainty range and a breakdown of how rainfall was distributed across seasons.
10. Navigate across the four tabs to explore historical trends, interactive scatter plots, the live prediction tool, and detailed model validation results.

Dashboard Tab	Contents
Overview	Summary statistics, annual yield trend chart, year-by-year data table with conditional colour formatting
Data Exploration	Interactive scatter plots comparing any variable against yield; ranked correlation chart for all 38 variables
Yield Predictor	Live prediction from the trained model; rainfall distribution chart; scenario comparison tool for saving and comparing multiple seasonal inputs
Model Performance	Actual vs predicted scatter plot; prediction error bar chart by year; variable importance chart; reliability assessment table

13. Data Sources and Acknowledgements

- Maize yield and area planted: Ministry of Agriculture and Livestock Development, Kenya.
- Weather data: NASA POWER Predictive Worldwide Energy Resources (power.larc.nasa.gov). Satellite-derived monthly climate variables for the Uasin Gishu County centroid coordinates, 2012–2023.
- Soil properties: Lomurut Research Station soil survey (2014), made available through Purdue University's soil data repository.
- Seed variety performance data: CIMMYT (International Maize and Wheat Improvement Center) and the Tegemeo Institute of Agricultural Policy and Development, Egerton University, Kenya.